

NL Question Q and Schema S

Node 1: Adaptive Router openai/gpt-oss-20b via Groq

COMPLEX

Node 2: Schema Explorer CHES Semantic Pruning + MCI Card/Range Profiling

S'_{enriched}

Node 3: SQL Generator Dual Candidate (DDL vs Markdown) + Sandbox Selection

y_0

Node 4: Resilient SQL Validator

Syntax/Empty/Null

Node 6: Targeted Corrector openai/gpt-oss-120b via Groq

SUCCESS

Corrected SQL y_{i+1} (Max 3)

Node 5: Output Aligner Column name & standard case alignment

Final SQL $\hat{y}$

SIMPLE (bypass)

AGENTSQL-ASYM LANGGRAPH STATE MACHINE

